

Qualitative Research Methods

Conceptual Foundations and Applications

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February 25, 2026

Analogy: Qualitative Study Designs

Studying qualitative designs is like choosing different camera lenses.

- Wide-angle lens captures cultural context (Ethnography).
- Zoom lens focuses deeply on one case (Case Study).
- Portrait lens captures lived experience (Phenomenology).

Real Research Example: Study Designs

- Case Study: Investigating one startup's AI adoption process.
- Ethnography: Studying developer culture in open-source communities.
- Phenomenology: Exploring experiences of first-year PhD students.

Qualitative Study Designs

- **Case Study**

Examines a bounded system in depth within its context.

- **Ethnography**

Studies shared patterns of behavior within a cultural group.

- **Phenomenology**

Investigates lived experiences to uncover core meanings.

- **Narrative Research**

Explores individual stories to understand personal meaning.

Analogy: Nature of Qualitative Research

Qualitative research is like exploring a forest without a fixed map.

- The path evolves as you walk.
- You observe patterns rather than measure distances.
- Interpretation shapes understanding.

Real Research Example: Nature

- Exploring how remote work reshapes team collaboration.
- Understanding student perceptions of AI-assisted learning.
- Studying emotional impact of job automation.

Nature of Qualitative Research

- **Exploratory**
Used when limited prior theory exists.
- **Interpretive**
Focuses on meanings rather than numerical measurement.
- **Contextual**
Emphasizes real-world settings.
- **Flexible Design**
Allows research questions to evolve.

Types of Qualitative Research

- **Grounded Theory**
Develops theory inductively from data.
- **Phenomenography**
Studies variations in understanding.
- **Action Research**
Combines intervention with inquiry.
- **Content Analysis**
Systematically interprets textual material.

Analogy: Coding Strategies

Coding is like organizing a large library.

- Books are grouped into themes.
- Related categories are connected.
- A main topic emerges across shelves.

Real Research Example: Coding

- Interview transcripts coded to identify themes like “burnout”.
- Forum posts categorized to detect trust in AI tools.
- Student feedback grouped into learning challenges.

Coding Strategies

- **Open Coding**
Identifies initial concepts from raw data.
- **Axial Coding**
Links categories to subcategories.
- **Selective Coding**
Integrates categories into central theory.
- **Thematic Coding**
Detects recurring themes across datasets.

Analogy: Data Collection

Collecting qualitative data is like gathering puzzle pieces.

- Interviews reveal personal insights.
- Observations capture behaviors.
- Documents provide historical evidence.

Real Research Example: Data Collection

- Conducting interviews with cybersecurity professionals.
- Observing DevOps practices in industry.
- Analyzing institutional policy documents.

Data Collection Methods

- **Interviews**

In-depth conversations to explore perceptions.

- **Observations**

Directly recording natural behaviors.

- **Focus Groups**

Group discussions revealing collective viewpoints.

- **Document Analysis**

Reviewing textual or digital artifacts.

Primary and Secondary Sources

- **Primary Data**
Collected firsthand from participants.
- **Field Notes**
Observational records during study.
- **Secondary Data**
Existing published or archival materials.
- **Literature Records**
Scholarly works supporting context.

Analogy: Data Validation

Validation is like cross-checking news from multiple sources.

- Different perspectives confirm accuracy.
- Participants verify interpretation.
- Documentation ensures transparency.

Real Research Example: Validation

- Triangulating interviews with observation logs.
- Member-checking PhD student interview transcripts.
- Peer review of coding categories.

Data Validation

- **Triangulation**

Uses multiple data sources to enhance credibility.

- **Member Checking**

Participants confirm interpretations.

- **Peer Debriefing**

External review strengthens reliability.

- **Audit Trail**

Maintains transparent research documentation.

Data Analysis in Qualitative Research

- **Thematic Analysis**
Identifies meaningful patterns in data.
- **Content Analysis**
Systematically categorizes textual content.
- **Narrative Analysis**
Examines stories for structure and meaning.
- **Discourse Analysis**
Analyzes language in social context.

Conclusion

- Qualitative research provides deep contextual insights.
- Method selection depends on research objectives.
- Rigorous validation enhances trustworthiness.
- Analytical rigor leads to theory generation.